

$$2.7.4 \quad A: X \rightarrow Y, \quad V \subseteq X \quad \Rightarrow \quad \dim \overline{AV} \leq \text{rank } A$$

subspace

image of V
under A .

1) The image of a subspace under a linear transformation is a subspace of the codomain (and of the range).

2) The dimension of a subspace is at most the dimension of its superspace.

$$3) \text{rank } A = \dim(\text{range } A)$$

$$4) \text{ Since } \dim AV \leq \dim(\text{range } A), \quad \dim AV \leq \text{rank } A$$

$$\text{rank } AB = \dim \text{range}(AB) = \dim(A \text{range}(B)) = \dim AV \leq \text{rank } A$$

$V = \text{range } B$

2-8-3

Convert $1, 1+t$ basis for P_1 to $1-t, 2t$ basis for P_1

Idea: first convert to standard basis,
then convert from standard basis to desired basis.

In standard basis, $1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $1+t = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

So to convert to standard basis, apply matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

In standard basis, $1-t = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $2t = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

So to convert from standard basis to this one,

apply matrix $\begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$

So composing yields: $\frac{1}{2} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix}$

Spot
check: $\begin{pmatrix} 1 & 1 \\ 1/2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$

And indeed, $2(1) + 3(1+t) = 3t+5 = \underset{8t-5t+5}{\overset{11}{5(1-t)+4(2t)}}$

II Characterize all matrices in $M_{2 \times 2}^{\mathbb{R}}$ that are equal to their own inverses.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Note $ad-bc = \pm 1$, since the determinants must be nonzero, be equal, and they are related as reciprocals:
 $x = \frac{1}{x} \Rightarrow x^2 = 1 \Rightarrow x^2 - 1 = 0 \Rightarrow x = \pm 1$.

So two cases.

$$\begin{aligned} \det A &= \det A^{-1} \\ &= \frac{1}{\det A} \end{aligned}$$

i) $\det A = 1$

$$\text{Then } \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \Rightarrow \begin{matrix} a=d \\ b=-b \\ c=-c \end{matrix} \Rightarrow \begin{matrix} a=d \\ b=0 \\ c=0 \end{matrix}$$

And the determinant condition further implies $ad-bc=1$,

$$\Rightarrow ad=1$$

$$\Rightarrow a^2=1$$

$$\Rightarrow a=\pm 1$$

So $a=d=\pm 1$, $b=c=0$. This yields

the two matrices: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

↑
identity

↑
180° rotation

ii) $\det A = -1$

Then $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -d & b \\ c & -a \end{pmatrix} \Rightarrow \begin{matrix} a = -d \text{ (or } -a = d) \\ b = b \\ c = c \end{matrix}$

And the determinant condition further implies $ad - bc = -1$
 $\Rightarrow -a^2 - bc = 1$

This yields the two-parameter family of matrices

$$\begin{pmatrix} a & b \\ \frac{a^2-1}{b} & -a \end{pmatrix} \text{ (if } b \neq 0 \text{)}$$

$$\begin{aligned} bc &= a^2 - 1 \\ c &= \frac{a^2 - 1}{b} \\ &\text{(if } b \neq 0 \text{)} \end{aligned}$$

plus $\begin{pmatrix} a & 0 \\ c & -a \end{pmatrix}$ with $a^2 = 1$

This is fine, but maybe not very satisfying.

Claim: all of these matrices are in fact similar to $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and so are reflections in some line through the origin in $\mathbb{R}^2$!

How to see this: We will reason in a coordinate-free way. First, suppose that A fixes some vector along its line — that is, A just acts as scalar multiplication by some factor. Then that factor must be 1 or -1, in order for $A^2 = I$

Now consider possibilities:

a) There are two linearly independent vectors that have this property. Then we can choose a basis for $\mathbb{R}^2$ consisting of these two ~~matrices~~ vectors. Writing down the matrices for the ± 1 possible scalings, we have the following possibilities, up to reordering the vectors:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

b) What happens if there is one vector v with this property, but not another that is linearly independent from it? Well take some w that completes to a basis v, w for $\mathbb{R}^2$.

We are supposing that Aw is neither w nor $-w$. Where does it go? Not to the line defined by kv , since then A would not be invertible. So w maps somewhere, Aw , which in turn maps back to w : $A(Aw) = A^2w = Iw = w$.

~~Consider the vector $w + Aw$.~~ Neither w nor Aw lies in kv , for if the latter were the case A would not be invertible (and it is: $A^{-1} = A$). Additionally, by assumption, Aw does not lie in kw . Then in the basis v, w , $Aw = \begin{pmatrix} a \\ b \end{pmatrix}$ where neither

a nor b equals 0. Consider the vector $w + Aw$.

~~Its~~ Its image under A is $A(w + Aw) = Aw + A^2w = Aw + w$, namely itself. This vector has coordinates $\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1+a \\ 1+b \end{pmatrix}$.

If $b \neq -1$, we are done, because we have found a vector linearly independent from v that is fixed (or negated) by A . Can $b = -1$? Sure,

but then in the ~~u~~ v, w basis the matrix for A is $\begin{pmatrix} \pm 1 & a \\ 0 & -1 \end{pmatrix}$, which is similar to $\begin{pmatrix} \pm 1 & 0 \\ 0 & -1 \end{pmatrix}$.

e) What if there are no (nonzero) vectors with this property? Well, that's not possible. Again, take any nonzero vector v and consider $v + Av$. We have $A(v + Av) = Av + A^2v = Av + v = v + Av$, so this vector is fixed by A . If $v + Av$ is nonzero, we are done, because contradiction. And if $v + Av = \vec{0}$, then $Av = -v$, and so A just scales v by -1 and so we again have found a vector with the desired property.

Supp III.

i) Sometimes. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 1 & 3 \end{pmatrix}$

but also $\begin{pmatrix} 2 & 1 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

vii) never. If there is a nontrivial solution, there is also a trivial solution (b/c homogeneous) and so the solution is not unique.

ii) Always. This is the Rank Theorem

iii) sometimes ~~$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$~~

nullity $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = 0$ and ~~nullity $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$~~

and nullity $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}^T = \text{nullity} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = 0$

but also nullity $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 2$

but nullity $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}^T = \text{nullity} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = 1$

iv) Sometimes

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is similar to itself.

But only to itself — eg $\begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}$ is not similar to $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(different determinants)

v) Sometimes $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is row equiv to an identity matrix

but $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is not (RR preserves rank)

vi) sometimes: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is similar to itself,

but $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ is invertible but not similar to $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

vii) Always. Invertible $\Rightarrow$ full rank $\Rightarrow$ LI columns $\Rightarrow$ row equiv to I.